Proposal for an Educational Game to Support Information Security Learning Among Brazil's Indigenous Peoples

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Abstract. Introduction: *This article presents the development of an educational game prototype for teaching information security in indigenous communities.* ***Objective:*** *The game follows an indigenous student who provides technical assistance in neighboring villages.* ***Methodology:*** *The prototype uses contextualized dialogues and mini-games to address preventive security practices and is being developed in Unity using C#.* ***Results:*** *The prototype is currently under development, with the initial implementation of the narrative, dialogues, and game mechanics.*
Keywords: *Educational Games; Information Security; Indigenous Communities; Human-Computer Interaction.*

Resumo. Introdução: *Este artigo apresenta o desenvolvimento de um protótipo de jogo educativo para o ensino de Segurança da Informação em comunidades indígenas.* ***Objetivo:*** *O jogo acompanha um estudante indígena que presta assistência técnica em aldeias vizinhas.* ***Metodologia:*** *O protótipo utiliza diálogos contextualizados e minijogos para abordar práticas preventivas de segurança, sendo desenvolvido na Unity com C#.* ***Resultados:*** *O protótipo encontra-se em desenvolvimento, com a implementação inicial da narrativa, dos diálogos e das mecânicas do jogo.*
Palavras-chave: *Jogos educativos; Segurança da Informação; Comunidades indígenas; Interação Humano-Computador.*

1. Introduction

Data from the Household ICT Survey indicate that approximately half of the indigenous population reported never having used a computer [CETIC.Br 2021]. This scenario of digital exclusion increases these groups vulnerability to cyberattacks and social engineering, since information security literacy has not kept pace with the rapid arrival of mobile devices in villages. According to [Werbach and Hunter 2012], gamification is not merely

about creating a game, but about applying game design elements to non-game contexts, such as employee training in cybersecurity practices.

Drawing on references to [Csikszentmihalyi 1990] Flow theory, which describes a mental state of total immersion and pleasure in an activity, where the person feels energized, focused, and loses track of time, achieving high performance and satisfaction, this occurs when there is a perfect balance between the challenges of the task and the individual's abilities, which is crucial for happiness and well-being.

Based on these scenarios, this study aims to develop a functional prototype of a serious game to assist in teaching cybersecurity and raising awareness among indigenous peoples, using simpler mini-games to demonstrate not how difficult this field is, but that, with a simple change, the problem can be solved or prevented. The proposed idea is based on the history of activities carried out in the Information Systems program at the Federal University of Western Pará -— Oriximiná Campus.

2. Related studies

The article by [de Souza et al. 2025] addresses information security through the story of a university student recruited to help the Resistance fight an organization that controls the world through information technology. Throughout the game, the player develops skills in logic, programming, computer networks, operating systems, and cybersecurity while solving problems in simulated real-world scenarios.

The serious game *picoCTF* [Chapman et al. 2023] is a gamified platform for challenge-based cybersecurity education, combining progression mechanics, challenges, and feedback. In contrast, *SherLOCKED* [Jaffray et al. 2021] adopts a detective theme, in which players investigate cybersecurity incidents through interactive storytelling, decision-making, and scenario-based problem-solving.

The work by [Scherb et al. 2023] presents a serious game focused on cyberattack simulation, allowing players to explore vulnerabilities and attack strategies in a controlled environment. Finally, *PeriHack* [Dillon and Arushi 2022] focuses on cybersecurity awareness through everyday digital scenarios, employing choice mechanics, consequences, and educational feedback to support learning.

3. Methodology

This study falls under the category of applied research, given its potential to generate knowledge through an application designed to solve a problem in the educational context of cybersecurity. According to [Gil 2010], applied research makes it possible to generate knowledge through concrete, everyday situations. The design of this educational game was divided into several key points:

Problem Statement: Indigenous people face significant challenges with the Portuguese language, which prevents them from acquiring sufficient knowledge in various fields, including cybersecurity. This obstacle was the primary motivation for developing this project.

Requirements Engineering: For this project to take shape, it was necessary to identify certain requirements, including:

- **Functional requirements:** contextualized narrative, offline mode, feedback system, and avatar selection (male or female);
- **Non-functional requirements:** lightweight performance, visual accessibility, data security, and low memory consumption;

Brainstorming: Throughout the project, we came up with several ideas on how to proceed, such as 2D, 3D, mini-games, quizzes, choice-based RPGs, or network defense simulations. We opted for a 2D style and simpler mini-games to demonstrate that solving cybersecurity challenges doesn't have to be a complex task;

Prototyping: At this stage, tools such as Unity, Visual Studio, RPG Maker, and C# were used to create the functional alpha prototype, which will be made available for testing by the target audience;

4. Results and discussion

4.1. Game development

The results obtained after completing all stages and the current programming were as follows: a 2D game designed to discuss information security methods with players. Regarding the plot, it tells the story of a recent graduate from UFOPA's Information Systems program who, after completing his specialization in cybersecurity, decided to return to his village and neighboring villages to implement a project in which he will solve problems and teach basic tips to the other tribal inhabitants.

The goal of this game is to teach and offer tips on information security, which is evident in the system of mini-games that do not present very difficult tasks, but rather more basic ones, since the objective is to make it easier for the player to understand and to show that solving a problem can be simple and easy to grasp, rather than a "huge challenge." The game itself features several screens, which are shown below along with their respective explanations:



Figure 1. Surroundings of the village



Figure 2. Arrival on the First village

Figure 1 shows the chosen character arriving at the outskirts of the first village in this section and includes a brief tutorial to help the player become familiar with the movement controls and how to interact with objects; specifically, the player must interact with the dry tree to trigger the transition to the next screen. On this screen (Figure 2), we'll arrive at the village where we'll interact with the game's first NPC so we can help with its current problem. To do this, we have two options: for players who want to enjoy the music and scenery, simply approaching her will trigger the dialogue. If you're in a hurry

or have an urgent matter, just click on her to start the dialogue—regardless of whether you've moved or not. It's worth noting the day-passing system, where the character starts in the morning and as they travel through the villages, the day progresses, creating a sense of immersion where you truly feel like you're visiting these places at different times of day.

4.2. Discussion

An analysis of the prototype's development suggests that the gamification strategy adopted aligns with the principles outlined by [Werbach and Hunter 2012], as the focus was not merely on entertainment but on applying game elements to address a real-world problem: the digital vulnerability of indigenous peoples. Unlike more complex or fantastical approaches, such as that seen in [de Souza et al. 2025] game "Cyber Resistance," which focuses on university settings and technological resistance, this project opted for a narrative of itinerant technical assistance.

A central point of the discussion is the application of the **Flow Theory** [Csikszentmihalyi 1990], which establishes a relationship between the level of challenge presented by the activity and the individual's skills. In this regard, the inclusion of low-complexity minigames, such as tic-tac-toe and basic puzzles, was an intentional design decision aimed at tailoring the challenges to the target audience's initial skill level. This strategy aims to avoid situations of frustration arising from overly complex challenges, as well as activities that are not very stimulating, thereby contributing to a more balanced interactive experience that is potentially conducive to player engagement.

5. Concluding remarks

The game addresses not only access to technology but also its critical and safe use, promoting awareness of practices to reduce data exposure and exploitation risks. The use of Indigenous characters and culturally relevant scenarios seeks to strengthen player identification and contextualize the content. The prototype was developed using *Unity* and C#, providing a suitable structure for its current implementation and future enhancements.

For future work, we plan to redesign the game using a participatory design methodology, seeking to build the game collaboratively with experts in both information security and education. Additionally, we intend to conduct a validation study with the target audience using the ARCS method, with the goal of assessing the level of learning and engagement in cybersecurity during gameplay. Based on player feedback, we intend to improve the game by implementing new modules.

Statement on the Use of AI

The authors declare that they used the ChatGPT tool exclusively for language refinement and LaTeX formatting. All content has been reviewed and validated by the authors, who assume full responsibility for the accuracy, originality, and final version of the manuscript.

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